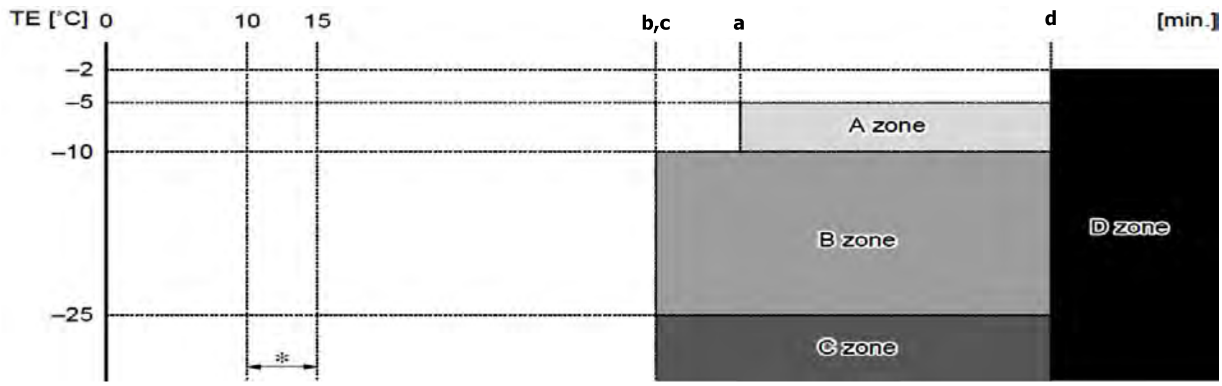
Item	Operation flow and applicable data, etc.	Description
7. Defrost control (Only in heating operation)	(This function removes frost adhered to the outdoor heat exchanger.) The temperature sensor of the outdoor heat exchanger (Te sensor) judges the frosting status of the outdoor heat exchanger and the defrost operation is performed with 4-way valve reverse defrost system.	The necessity of defrost operation is detected by the outdoor heat exchanger temperature. The conditions to detect the necessity of defrost operation differ in A, B, or C zone each. (Table 1)

Start of heating operation



* The minimum TE value and To value between 10 and 15 minutes after heating operation has started are stored in memory as TE0 and TO0, respectively.

Table 1

Defrost zone	In normal To	In abnormal To
A zone	TE0-TE≥3°C & SH-SH0≤2	(TE0-TE)-(TO0-TO)≥3°C & SH-SH0≤2
B zone	TE0-TE≥2°C & SH-SH0≤2	(TE0-TE)-(TO0-TO)≥2°C & SH-SH0≤2
C zone	TE≤ -25°C & SH-SH0≤2	
D zone	More than 90 minutes accumulate heating operation time condition TE≤ -2°C	

Table 2

Heating operation (time)	Model			
	07k	10k	13k	16k
a	43		51	
b	39		49	
c	31			
d	90			

<Defrost operation>

- Defrost operation in A to C zones

1) Stop operation of the compressor for 40 seconds.
2) Invert (OFF) 4-way valve 40 seconds after stop of the compressor.
3) The outdoor fan stops at the same time when the compressor stops.
4) When temperature of the indoor heat exchanger becomes 38°C or lower, stop the indoor fan.

<Finish of defrost operation>

- Returning conditions from defrost operation to heating operation

1) Temperature of outdoor heat exchanger rises to +8°C or higher for 3 seconds.
2) Temperature of outdoor heat exchanger is kept at +7°C or higher for 60 seconds.
3) Defrost operation continues for 10 minutes.

<Returning from defrost operation>

1) Stop operation of the compressor for approx. 40 seconds.
2) Invert (ON) 4-way valve approx. 30 seconds after stop of the compressor.
3) The outdoor fan starts rotating at the same time when the compressor starts.